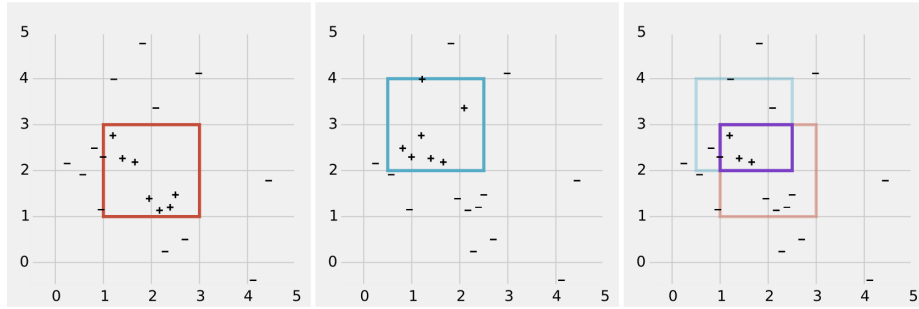


1 VC Dimension

Recall that we call a set of points *shattered* by a class of functions H if all possible $\{-1, +1\}$ labelings of the points can be produced by some function in H . The *Vapnik-Chervonenkis* (VC) dimension is the size of the largest set of points that can be shattered by the hypothesis space.

In this problem, we will explore the hypothesis space where each hypothesis is a combination of two simpler hypotheses. More precisely, given two hypotheses h_1 and h_2 , we define $h = h_1 \cap h_2$ as a new hypothesis that labels an example $+1$ only if both h_1 and h_2 give the label $+1$, otherwise, it is labeled -1 . We can extend this to *sets* of hypotheses: given two sets of hypotheses H_1 and H_2 , define $H^* = \{h_1 \cap h_2 : h_1 \in H_1, h_2 \in H_2\}$ as the set of all intersections of hypothesis pairs from the two classes H_1 and H_2 .

As an example, let H_1 be the set of classifiers in $\mathbb{R}$ that assigns the label $+1$ if the example is larger than some threshold a . Let H_2 be the set of classifiers in $\mathbb{R}$ that assigns the label $+1$ if the example is smaller than some threshold b . Then H^* would be the set of all intervals (a, b) in $\mathbb{R}$ that assigns $+1$ if the example is inside the interval. Another example is when H_1 and H_2 is the set of all (axis-aligned) squares in $\mathbb{R}^2$, H^* is the set of all axis-aligned rectangles. This example is illustrated below. On the left, we have a single square classifier h_1 ; in the middle we again have a square classifier h_2 ; and on the right, we have $h_1 \cap h_2$, which is a rectangle classifier.

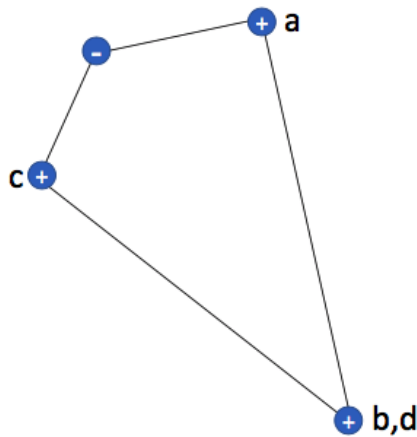


Keep in mind that these are only examples. We are looking for results that can apply generally to any pair of hypotheses classes.

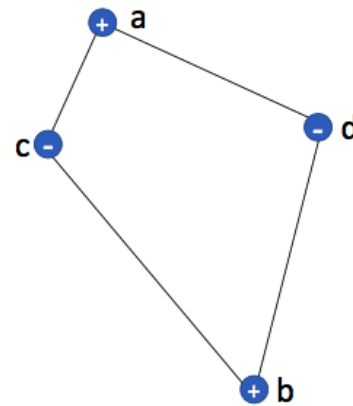
1. Suppose that the *shattering coefficient* of H_1 is $H_1[n]$ (i.e. the maximum number of ways that the hypothesis space H_1 can label a set of n points is $H_1[n]$). Similarly, suppose that the shattering coefficient of H_2 is $H_2[n]$. Show that $H^*[n] \leq H_1[n]H_2[n]$.

Solution: Let S be any set of n points. Let $\mathcal{L}_1$ be the set of all labelings of S by hypotheses in H_1 (so $|\mathcal{L}_1| \leq H_1[n]$). So for any $\ell_1 \in \mathcal{L}_1$, there is some hypothesis $h_1 \in H_1$ that produces the labeling ℓ_1 on S . Similarly, define $\mathcal{L}_2$ (so $|\mathcal{L}_2| \leq H_2[n]$) and so, for any $\ell_2 \in \mathcal{L}_2$, there is some hypothesis $h_2 \in H_2$ that produces the labeling ℓ_2 on S . Let $\mathcal{L}^*$ be the set of all labelings of S by hypotheses in H^* . So, for any $\ell^* \in \mathcal{L}^*$ there is some hypothesis $h^* \in H^*$ that produces the labeling ℓ^* on S . Notice that for any $h^* \in H^*$ there must exist $h_1 \in H_1$ and $h_2 \in H_2$ such that $h^* = h_1 \cap h_2$. Therefore, for any $\ell^* \in \mathcal{L}^*$ there must exist $\ell_1 \in \mathcal{L}_1$ and $\ell_2 \in \mathcal{L}_2$ such that ℓ^* is determined by ℓ_1 and ℓ_2 (on each point $x \in S$, $\ell^*(x) = \min[\ell_1(x), \ell_2(x)]$). Since there are at most $|\mathcal{L}_1|$ choices of ℓ_1 and at most $|\mathcal{L}_2|$ choices of ℓ_2 , we see that $|\mathcal{L}^*| \leq |\mathcal{L}_1||\mathcal{L}_2|$. Finally, we conclude that $H^*[n] = \max_S |\mathcal{L}^*| \leq \max_S |\mathcal{L}_1||\mathcal{L}_2| \leq \max_S |\mathcal{L}_1| \max_S |\mathcal{L}_2| = H_1[n]H_2[n]$.

2. For each one of the following function classes, find the VC dimension. State your reasoning.
 - i. Half spaces in $\mathbb{R}$, where examples on one side of the boundary are labeled $+1$, and examples on the other side are labeled -1 .



Case 1: the labeling above
is not achievable



Case 2: the labeling above
is not achievable

Solution: The VC dimension is at least 2, since the classifier can shatter any set of two distinct points in $\mathbb{R}$. However, this classifier cannot shatter a set of three points, since they cannot be labeled $+1, -1, +1$ from left to right. We conclude that the VC dimension is 2

- ii. Half spaces in $\mathbb{R}^2$, where examples on one side of the line are labeled $+1$, and examples on the other side are labeled -1 .

Solution: The VC dimension is at least 3, since the classifier can shatter any set of three non-collinear distinct points in $\mathbb{R}^2$. For sets of 4 points, the points either are collinear, form a triangle with an interior point, or form a quadrilateral. If they are collinear, then a labeling similar to the one in the previous part cannot be achieved. If they form a triangle, then a labeling of $+1$ on its vertices and of -1 on the interior point cannot be achieved. If they form a quadrilateral, then a labeling of $+1$ on one set of diagonals and -1 on the other cannot be achieved. Thus, the VC dimension is 3

- iii. Axis-aligned squares in $\mathbb{R}^2$, where points are labeled $+1$ inside the square, and -1 outside (as in the illustrations above).

Solution: The VC dimension is at least 3 since we can arrange 3 points to be at locations $\{(-1, 0), (0, 1), (1, 0)\}$ and axis-aligned squares can produce all possible labelings on these three points. However, no set of 4 points can be shattered. In particular, given 4 points, let a denote the topmost point, b denote the bottommost point, c denote the leftmost point, and d denote the rightmost point. Case 1: if any point was double-counted (e.g., if $b = d$) then there must be some remaining point, and it is not possible to label the remaining point as -1 and to label a, b, c, d as $+1$ (see Case 1 in figure). Case 2: no point was double-counted. In this case, if the vertical distance between a and b is greater than or equal to the horizontal distance between c and d then it is not possible to label a and b as $+1$ and to label c and d as -1 (see Case 2 in figure); likewise, if the horizontal distance between c and d is greater than or equal to the vertical distance between a and b then it is not possible to label c and d as $+1$ and to label a and b as -1 .

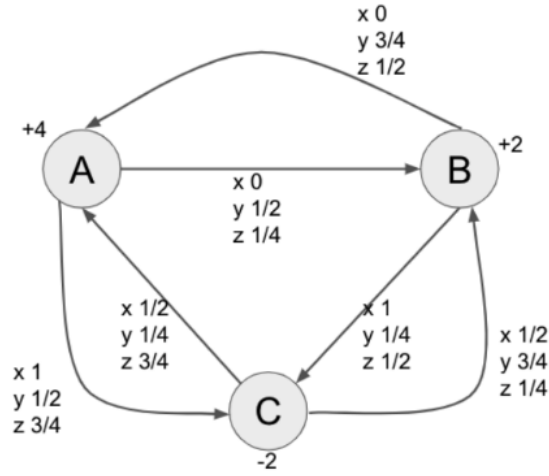


Figure 1: MDP

2 Reinforcement Learning

Consider the following Markov Decision Process (MDP) with discount factor $\gamma = 0.5$. Upper case letters A, B, C represent states, arcs represent state transitions; lower case letters x, y, z represent actions; signed integers represent rewards; and fractions represent transition probabilities. Discount factor γ is 0.5. Consider the policy π that takes action x in any state (i.e $\pi_i(s) = x$ for all $s \in \{A, B, C\}$). Compute the value function of this policy (i.e compute $V^\pi(A)$, $V^\pi(B)$, $V^\pi(C)$).

$$V^\pi(A) = 11/3, V^\pi(B) = 5/3, V^\pi(C) = -2/3$$

3 Support Vector Machines

You are given a dataset $\{x_i, y_i\}_{i=1}^n$ of n points for a binary classification task. Here, x_i 's are Euclidean vectors and $y_i \in \{0, 1\}$. You wish to use a soft-margin SVM to classify the points. Recall that the primal problem can be written as follows:

$$\begin{aligned} \min_{w, b, \xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t. } \quad & \forall i, y_i(\langle w, x_i \rangle + b) \geq 1 - \xi_i \text{ and } \xi_i \geq 0 \end{aligned} \quad (1)$$

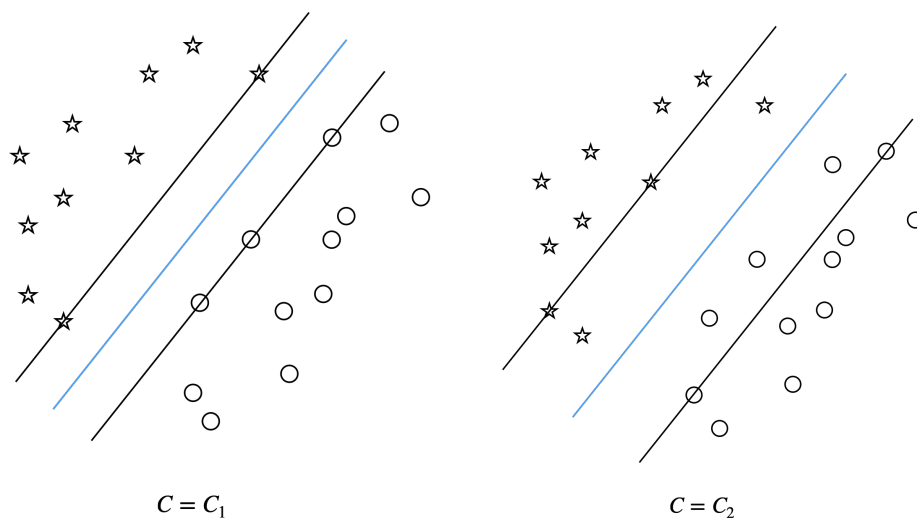
1. Generally speaking, what is the *main reason* for solving the dual of the optimization problem?

The dual problem can be written in terms of the inner products, allowing us to apply the kernel trick.

2. Let $\{\alpha_i\}_i$ denote the dual variables corresponding to the $y_i(\langle w, x_i \rangle + b) \geq 1 - \xi_i$ constraints. Suppose we solved the dual problem with $C = 5$ using a dataset of $n = 10$ points and found that the optimal α_i values were $[5, 4.99, 1.01, 1, 0.99, 0.01, 0, 0, 0, 0]$. List all the support vectors. You may list the indices of the points which correspond to the support vectors.

The first 6 points.

3. The two figures below show the decision boundary and the margin of an SVM trained on some labeled data with regularization parameters $C = C_1$ (left) and $C = C_2$ (right). The blue line represents the decision boundary, and the black lines from either side represent the size of the margin from the decision boundary. We have indicated the two classes using circles ($\circ$) and stars ($\star$).



- (a) Circle the points corresponding to the support vectors in *both* figures.

Points on and in the black lines.

- (b) Of the following five statements about C_1 and C_2 , which *one* is the most accurate based on the figures above. Circle the right answer.

(i) $C_1 < C_2$ (ii) $C_1 \leq C_2$ (iii) $C_1 = C_2$ (iv) $C_1 \geq C_2$ (v) $C_1 > C_2$

Ans: (v) $C_1 > C_2$